

Chair of Thermodynamics of Mobile Energy Conversion Systems (TME)

Prof. Marco Günther, RWTH Aachen University

Forckenbeckstrasse 4, 52074 Aachen

10 July 2026

Application: Research Assistant/Associate (PhD), Fuel Cell Systems (Ref. V000011113 SOFC / V000011114 PEM CFD & Water Transport / V000011115 PEM Aging)

Dear Prof. Günther,

I am applying for the Research Assistant/Associate PhD positions in fuel-cell systems at TME, across all three open topics (SOFC systems, PEM CFD and water transport, and PEM aging). For the last two months at DLR I have been designing the cathode-side control architecture for our aircraft fuel-cell system: a full 0D lumped-parameter plant model of the cathode loop plus feed-forward, PI and decoupler control for cathode operation and back-pressure-valve protection, now being tuned against real system measurement data in MATLAB/Simulink. That system-level modeling and control work sits on top of two years of hands-on PEM fuel-cell experience across scales, from a 1 kW student-competition stack to a 100 kW, 1.5 t FCEV pickup truck at Robert Bosch to the 150-200 kW aircraft powertrain at DLR.

On the experimental side, I am leading this season's fuel-cell system testing, integration and troubleshooting at Ecogenium on our newly built water-cooled stack, building on the standalone test bench I engineered and commissioned there, where I ran purging, humidity-cycling and polarisation-curve campaigns and mapped the stack's thermal operating window. I correlate that test data against my own models the same way I did at Bosch, where I validated a Simulink operating-strategy model against measurement data and built post-processing and KPI analysis covering power distribution, efficiency and degradation. That measurement-correlated, model-based workflow maps directly onto validating CFD and aging models at the single-cell and stack test bench, and the system-architecture and control principles I work with (balance-of-plant, control loops, operating-strategy optimization) carry over from PEM to SOFC, even though my hands-on stack experience so far is PEM.

A PhD is genuinely half of what I am weighing for what comes next, alongside industry fuel-cell roles, and continuing at RWTH under Prof. Pischinger's chair, in the city where I already live and work, is an easy decision. I would be glad to contribute to whichever of the three topics suits the group's priorities best: my cathode-control and modeling background speaks most directly to the SOFC systems position, and my test-bench and measurement-correlation experience to the CFD-validation and aging positions. I expect to submit my Master's thesis at DLR later this year and could start soon after. My English is fluent and my German is at A2-B1.

Yours sincerely,

Vallabh Pataneni